

COLBY SCHULZ

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Product-forward engineering leader who turns ambiguous ideas into applied AI software. Led an enterprise innovation pipeline through 12 zero-to-one alphas resulting in 4 funded betas and 1 concept scaled to global release. A former construction engineer who rebuilt his own professional identity and is eager to help others do the same as AI redefines theirs.

EXPERIENCE

Lead Software Engineer II, Deloitte Digital | Jan 2024 – Present

Innovation program lead for a global life sciences client | New York, NY

- Led engineering for 2+ years after helping win the engagement, which became the blueprint for a new forward-deployed engineering practice within the firm.
- Ran alphas and betas off a riskiest-assumption matrix built with stakeholders in interactive kickoff sessions. Assumptions defined the build rather than a feature list. First stakeholder demo the week after kickoff.
- Partnered with product on value cases and intake scoring. Carried the engineering angle from idea submission through completion and documentation handoff.
- Compressed alpha cycles from 4 weeks to 1 week as AI-assisted development matured. The binding constraint moved from engineering capacity to stakeholder alignment.

Selected alphas and betas

- **Voice-first IT troubleshooting assistant** (alpha to funded beta): Replaced an opaque ticket-and-wait process for meeting-room failures with a conversational agent that resolved issues live and escalated to a populated ServiceNow ticket, testing whether structured ticket data could be generated from open conversation. On-site testing reshaped it twice.
- **Computer-vision gowning compliance** (alpha to funded beta): Hybrid ML and generative AI system verifying PPE before operators entered manufacturing areas, built as a job aid rather than an evaluation tool. Ran structured experiments against accuracy, latency, user disruption, and ease of setup across facilities, then validated across skin tones, genders, and heights with testers recruited on site.
- **Smart-glasses evaluation**: Prototyped four concepts, then killed all four after testing assumptions around enterprise connection interface, eye strain, hands-free form factor, interface density, PPE affordances, and text readability.
 - *Remote assist*: wearer's view streamed to a remote expert with real-time annotations.
 - *IT tech assist*: heads-up display of tickets, SOPs, and live AI troubleshooting.
 - *Lab assist*: label reading and experiment documentation.
 - *Warehouse assist*: pallet identification for pickers.
- **Embodied AI probe** (4 weeks): Captured human demonstrations for robot training through Apple Vision Pro teleoperation and Isaac Lab. Stood up simulation infrastructure that had never run in the enterprise environment.

Software Engineer → Principal Engineer, Giant Machines | Jan 2017 – Jan 2024

Zero-to-one product consultancy, acquired by Deloitte Digital | New York, NY

- Represented engineering in acquisition discussions with Deloitte as one of a small group selected by leadership.
- Grew from IC to technical leadership over a 7-year tenure building digital products across zero-to-one engagements in healthcare, financial services, energy, and automotive.

Selected projects

- **State Street Verus**: Front end and product on a mobile-first AI research tool that connected breaking news to portfolio holdings and surfaced indirect second- and third-order exposures.
- **Elovee**: React Native iPad application delivering a synthetic familiar presence to seniors with dementia. Built with memory care experts and clinical researchers.
- **Major US utility**: iPad field application for cathodic-protection technicians monitoring buried pipeline anodes. Learned the survey workflow firsthand through on-site discovery at their training facilities.

Construction Engineer, T.Y. Lin International

Oct 2011 – Jul 2016

- Managed construction on major transportation infrastructure, including highways and one of the first US high-speed rail lines.
- Verified work to spec, measured completed work for payment, and prepared audit documentation as a frontline professional.

SKILLS

Applied AI: Coding agents, LLM application architecture, experiment design & evals, RAG pipelines, embeddings & vector search, realtime voice, vision-model evaluation, prompt engineering & guardrails

Engineering: TypeScript, React, React Native, Node, Python, FastAPI, PostgreSQL, Docker, Kubernetes, CI/CD, Azure, enterprise integration (ServiceNow, SSO/Azure AD)

Product: Zero-to-one prototyping, discovery/kickoff/assumption mapping, user testing, client demos, RFP pursuit, team leadership & mentoring

EDUCATION & CERTIFICATIONS

Claude Certified Architect – Foundations (CCA-F), Anthropic, 2026

B.S. Civil Engineering, University of Illinois Urbana-Champaign, 2011 | **Full Stack Software Development**, Flatiron School, 2016